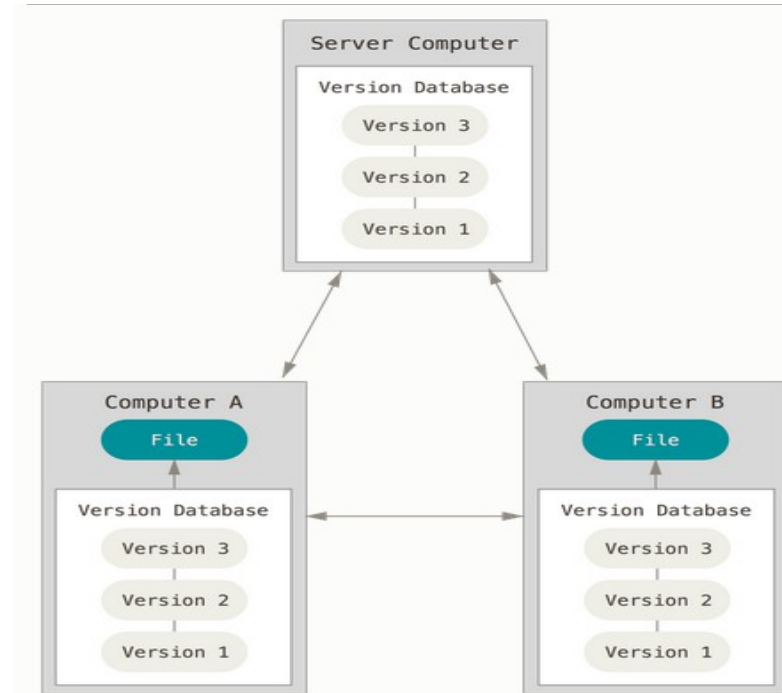
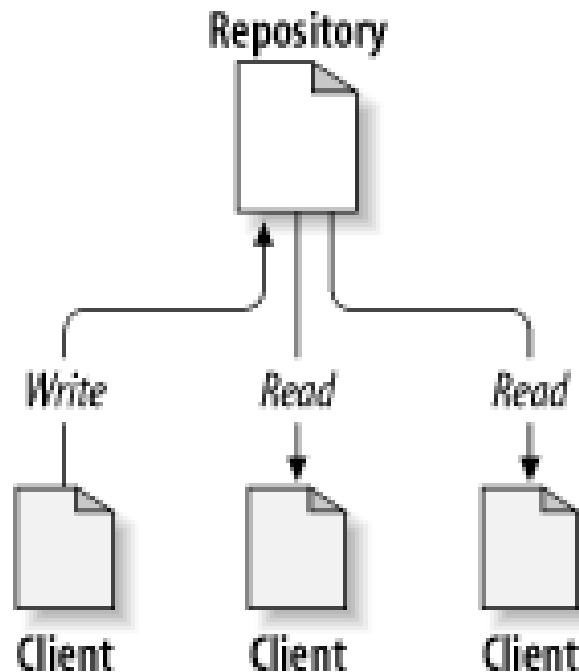


Version Control

Version Control

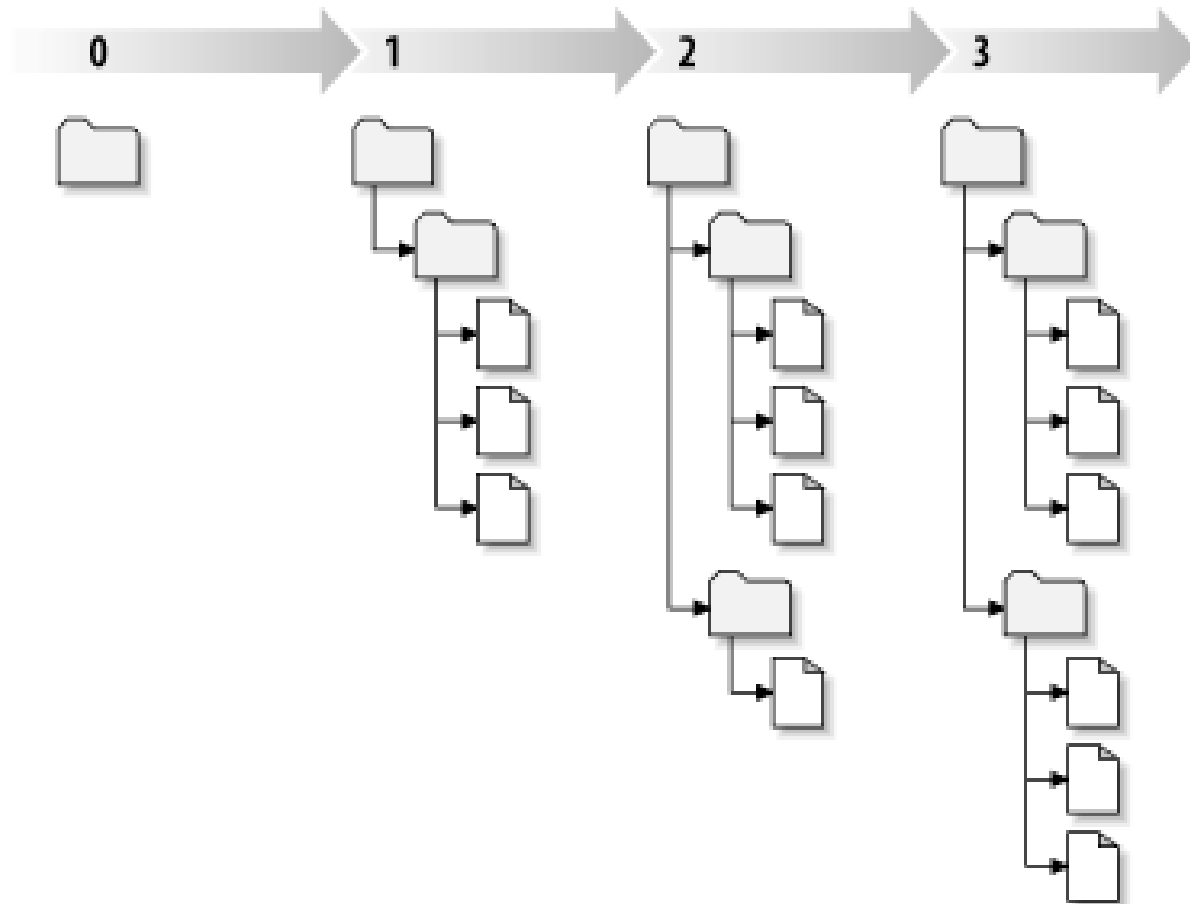
- Identify and store artifacts in secure repository
- Manage artifacts versions
- Keep track of changes made to artifacts to facilitate audits
- Support concurrent changes to artifacts while ensuring consistency
- Ensure reproducibility of builds

Repository

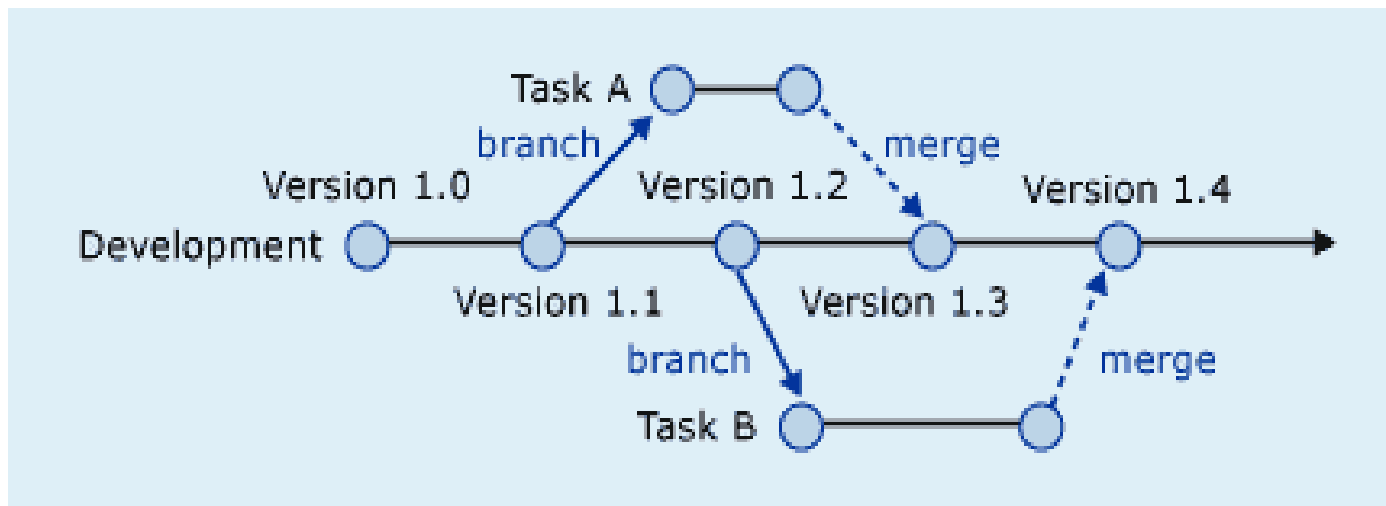


- Centralized or distributed repositories
 - Clients maintain their working copies, make changes locally and submit them to the repository where those are integrated
 - Submission results in a new revision

Revision



Branches



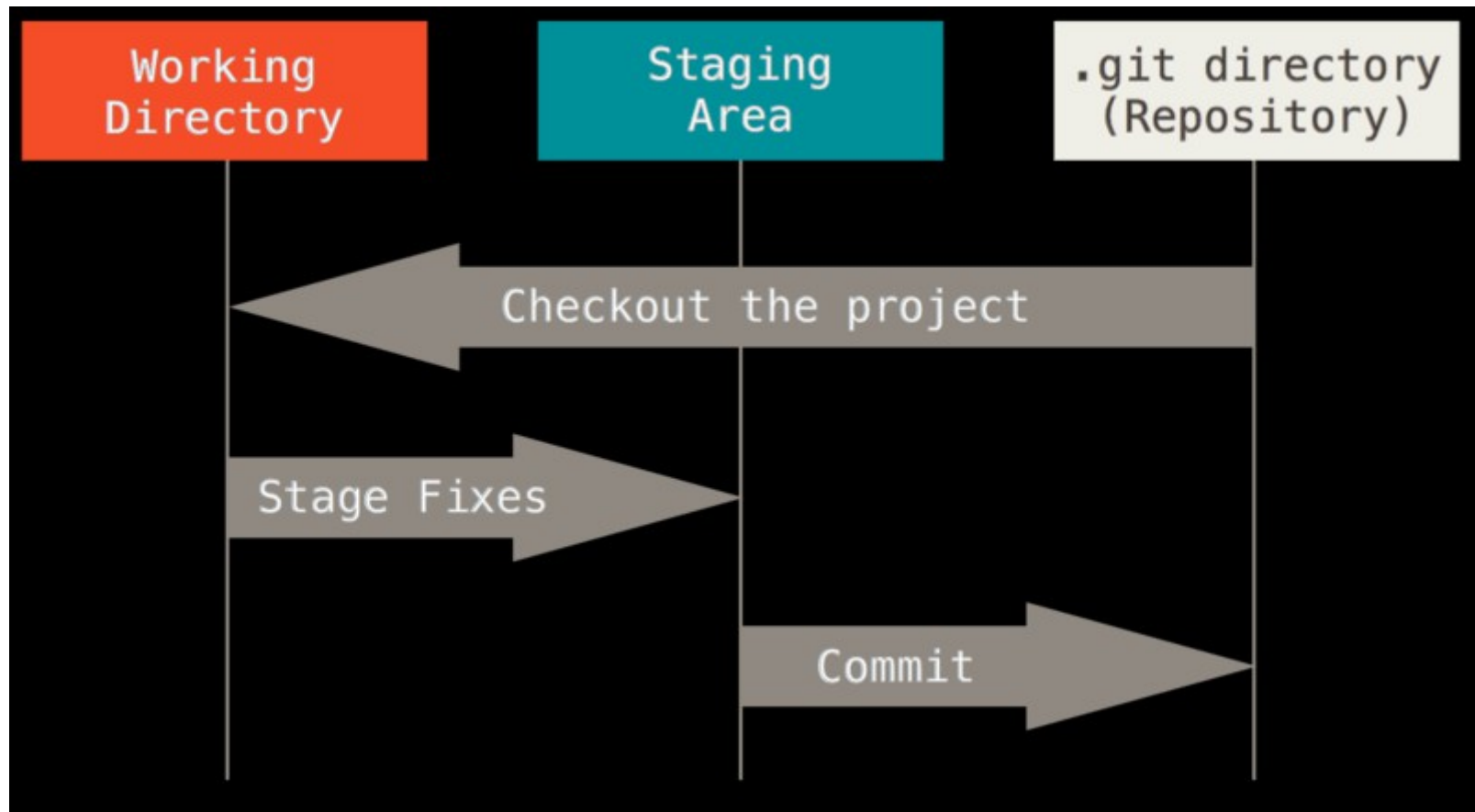
Basic concepts

- Create repository with proper access control
- Checkout / Clone to get working copies
- Update repository through basic work cycle
- Version / Label / Tag important milestones
- Create and maintain branches as necessary

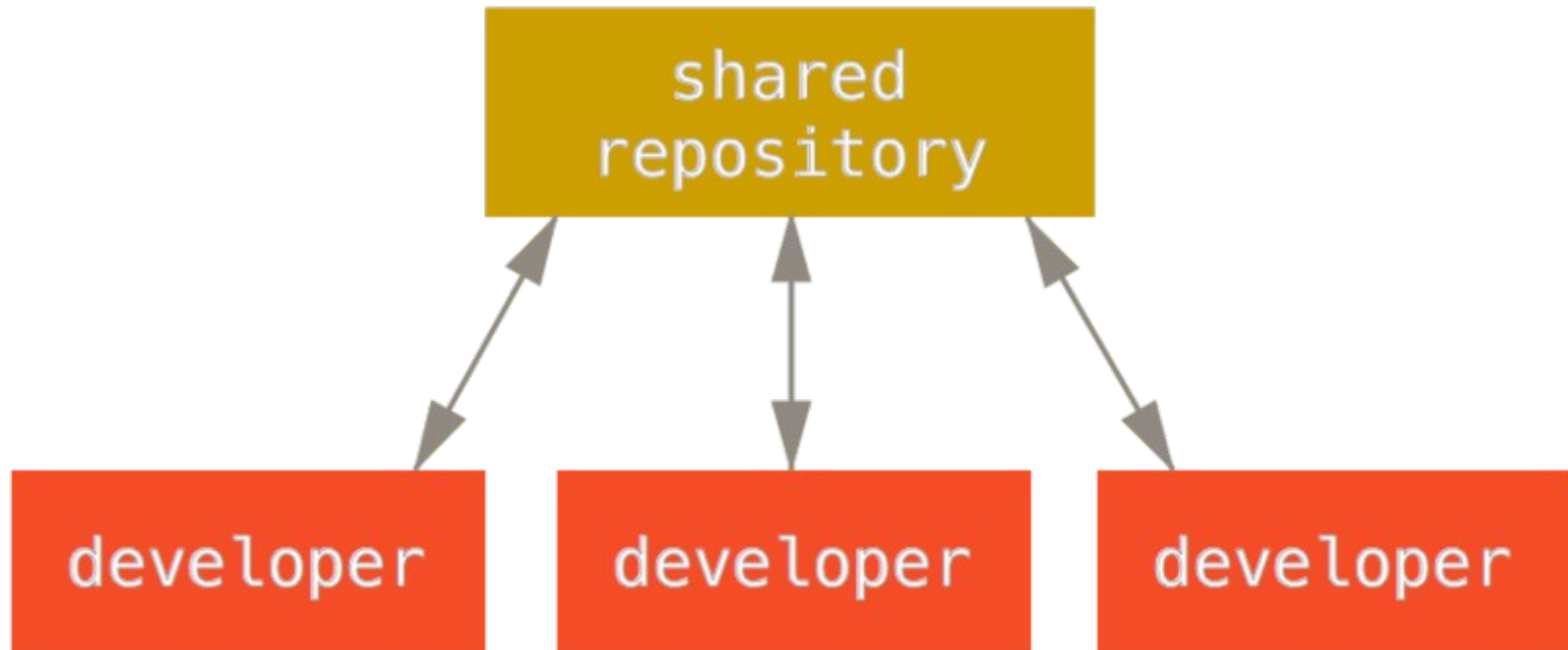
Basic work cycle using Git

- **Create / Clone repository**
 - `git init`
 - `git clone`
- **Make changes**
 - `git add`
 - `git rm`
 - `git reset`
 - `git checkout`
- **Examine changes / history**
 - `git status`
 - `git log`
 - `git diff`
- **Commit your changes**
 - `git commit`
 - `git push`
- **Branching, merging and tagging**
 - `git branch`
 - `git merge`
 - `git fetch / pull`
 - `git tag`

States

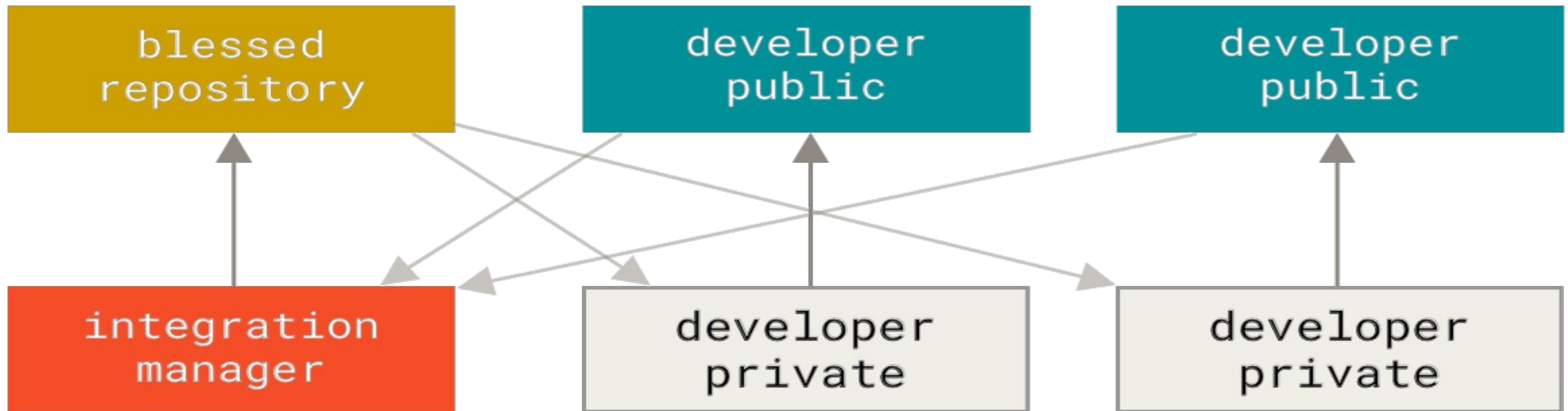


Centralized Workflow



Better maintain centralized workflows using **Feature-Branching** and **Pre-Integration Reviews** (Pull-Requests)

Integration-Manager



Works similar to a Fork

Dictator and Lieutenants Workflow

